

Eventstudy Analysis for Displacement Events

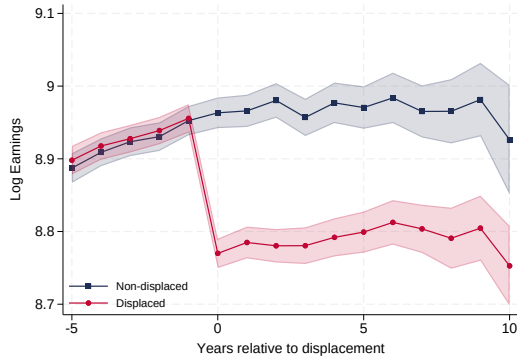
1 Comparing Eventstudy Specifications

Next, we compare the eventstudy estimates across different specifications.

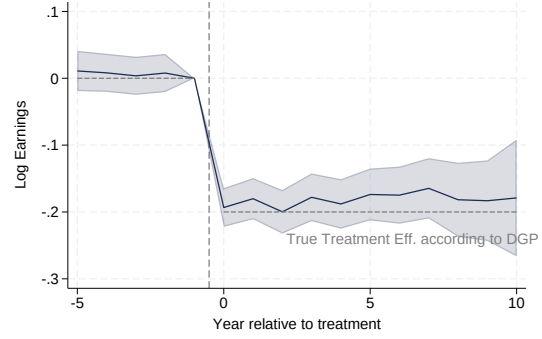
- Raw Means (always good to compare to the eventstudy estimates)
- OLS - Rel. year effects, no person FE
- JLS Specification: Cal. year and person FE - since we are not controlling for relative year effects, the estimates are not consistent
- Rel. year effects, person FE
- Relative year, calendar year and person FE
- Controlling for fully interacted calendar and relative year effects, as well as person FE

The matching design does the heavy lifting, that's why these specifications yield very similar results. Standard errors are clustered at the matched pair level (Abadie and Spiess, 2022).

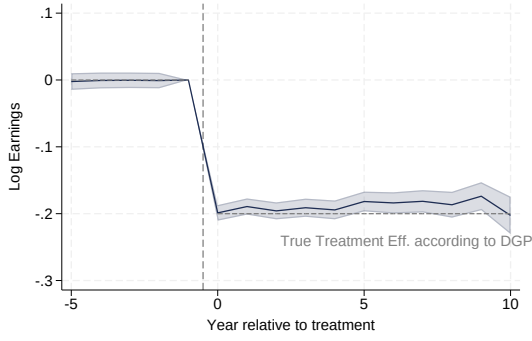
Figure 1: Comparing Eventstudy Specifications



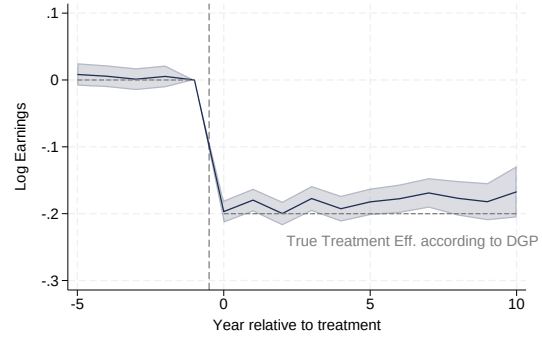
(a) Raw Means



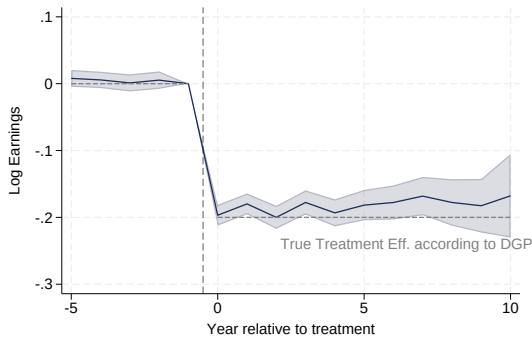
(b) Rel. year effects, no person FE



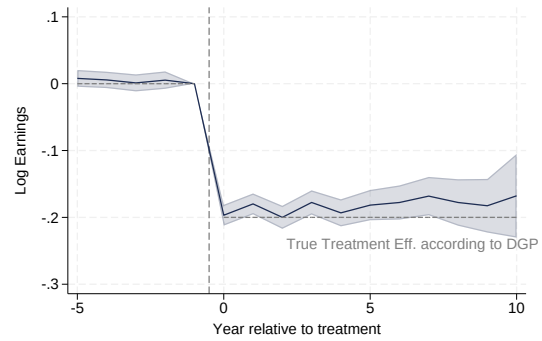
(c) Cal. year and person FE - does not work!



(d) Rel. year effects, person FE



(e) Rel. year, cal. year and person FE



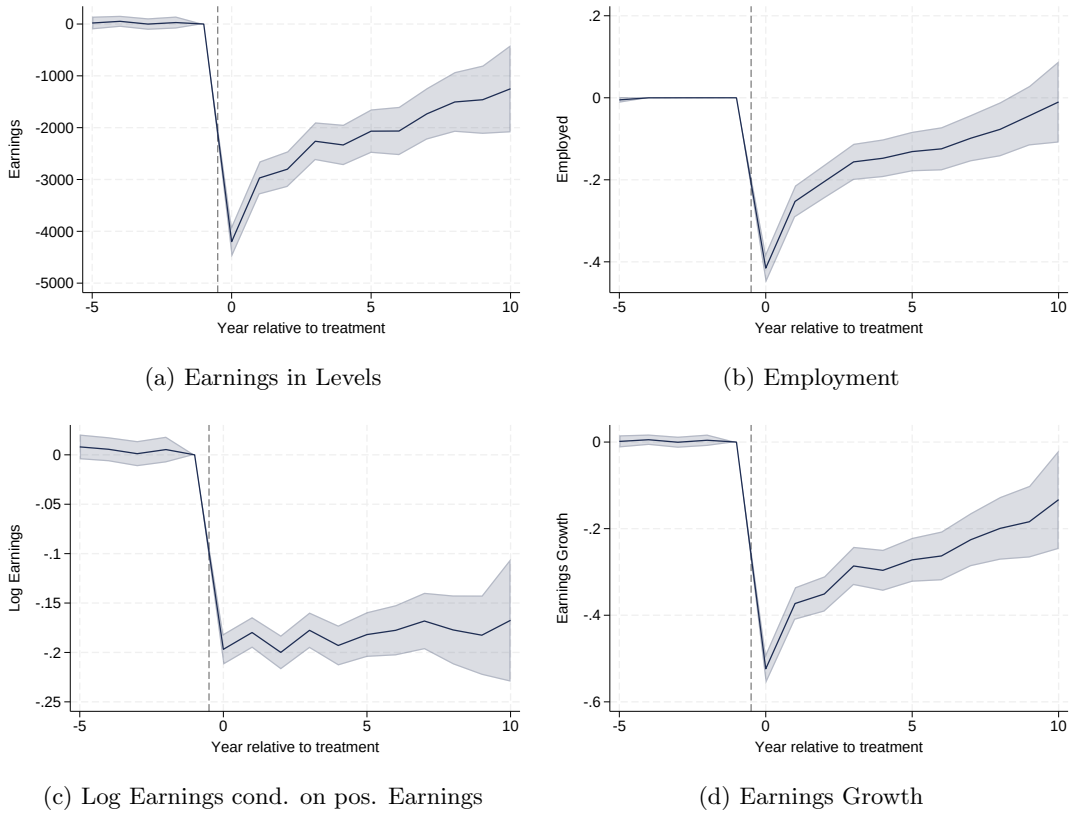
(f) Rel. year $\times$ cal. year and person FE

2 Different Outcomes

Next, we create eventstudy estimates for different outcomes.

- Earnings in Levels
- Earnings Growth
- Log Earnings conditional on positive earnings
- Employment

Figure 2: Event Study Estimates for Different Outcomes



3 Different Outcomes

Next, we create eventstudy estimates for different outcomes.

- Scale Raw Mean Difference by Control Mean:
 - Outcome: $\frac{\bar{y}_t^D - \bar{y}_t^{ND}}{\bar{y}_t^{ND}}$
- Scale Regression Coefficient from level regression by Control Mean:
 - Outcome: y_{it}
 - Regression: $y_{it} = \alpha_i + \gamma_t + \sum_k \beta_k D_{it}^k + \varepsilon_{it}$
 - Scale: $\frac{\hat{\beta}_t}{E[y_{it}|X_{it}, Non-Displaced]}$
- Poisson QMLE:
 - Outcome: `it(sameasApproaches1and2)`
 - Regression model: $E[y_{it}|X_{it}] = \exp(\alpha_i + \gamma_t + \sum_k \beta_k D_{it}^k)$,
 - Scale: $\exp(\hat{\beta}_t) - 1$
- Earnings Growth in percent:
 - Outcome: $\Delta y_{i,t} = \frac{y_{i,t} - y_{i,-1}}{y_{i,-1}}$
 - Regression: $\Delta y_{i,t} = \alpha_i + \gamma_t + \sum_k \beta_k D_{it}^k + \varepsilon_{it}$
 - Scale: β_t

Discussion: Comparing Approaches for Percentage Interpretation

Each of these approaches expresses earnings losses as a percentage, but they differ in their assumptions, what they estimate, and how they handle zero earnings.

Scaling by the control group mean (Approaches 1 and 2). With a matching design, the control group mean is a valid estimate of the counterfactual $E[Y(0)|D = 1]$, so dividing the treatment–control difference by the control mean yields a proper estimate of the ATT in percentage terms: $ATT/E[Y(0)|D = 1]$. Approach 1 (raw mean difference) is fully nonparametric—transparent and easy to compute. Approach 2 adds person fixed effects and controls, which may improve precision and absorb remaining imbalance after matching, but should yield similar point estimates if matching quality is high. Both approaches handle zero earnings naturally since they operate in levels. A minor disadvantage is that confidence intervals from dividing $\hat{\beta}_t$ by $\bar{y}_t^{ND}$ are approximate—strictly, one would need the delta method or bootstrap to account for estimation error in the denominator. This is the classic approach used in the job displacement literature (Jacobson, LaLonde, and Sullivan, 1993, *AER*).

Poisson QMLE (Approach 3). Like Approaches 1 and 2, Poisson QMLE uses earnings in levels y_{it} as the outcome and estimates essentially the same parameter—the percentage treatment effect—but does so directly via the multiplicative model $E[y_{it}|X] = \exp(\alpha_i + \gamma_t + \sum_k \beta_k D_{it}^k)$, so $\exp(\hat{\beta}_t) - 1$ gives the percentage effect without requiring post-estimation rescaling. Key properties:

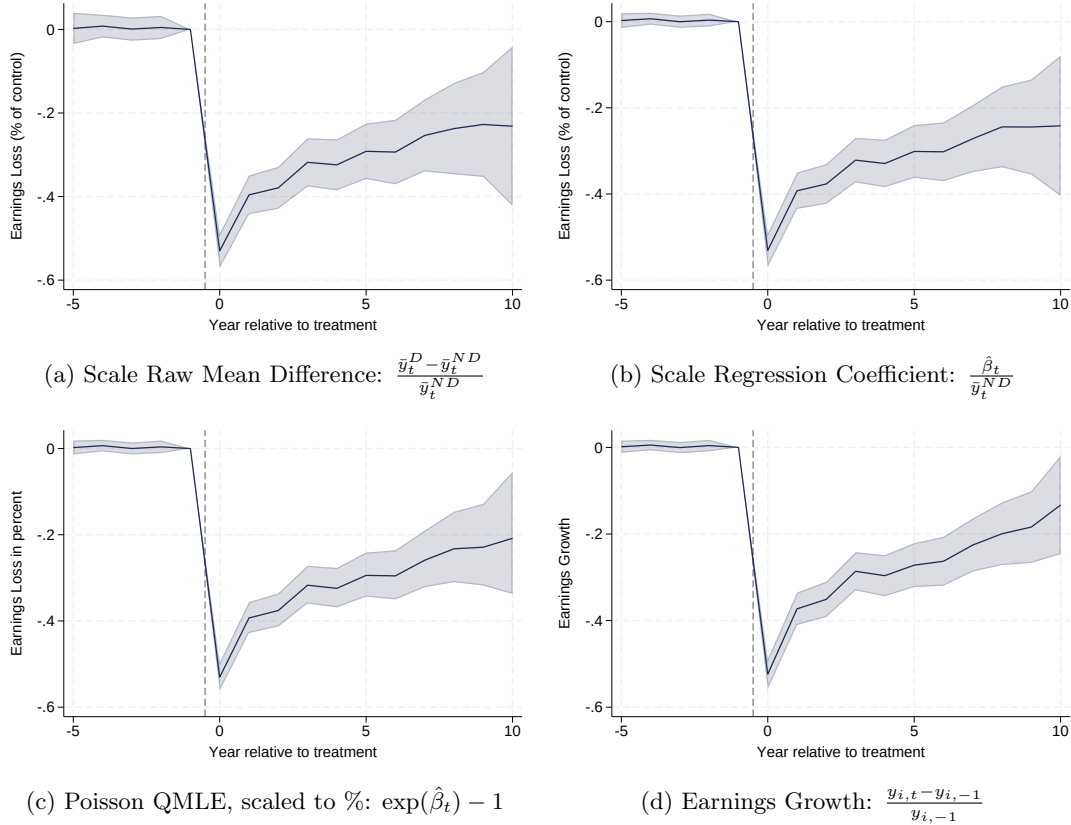
- *Handles zeros naturally:* Unlike $\log(y)$, Poisson PML is well-defined for $y = 0$ since it only requires correct specification of the conditional mean, not that y follows a Poisson distribution (Santos Silva and Tenreiro, 2006, *REStat*; Gourieroux, Monfort, and Trognon, 1984, *Econometrica*). High-dimensional fixed effects can be included via `ppmlhdfc` (Correia, Guimarães, and Zylkin, 2020, *Stata Journal*).

- *Functional form difference vs. OLS:* OLS in levels assumes **additive separability**: $E[y|i, t, D] = \alpha_i + \gamma_t + \beta D$ —the treatment effect is the same *absolute* amount for everyone. Poisson assumes **multiplicative separability**: $E[y|i, t, D] = \exp(\alpha_i) \cdot \exp(\gamma_t) \cdot \exp(\beta D)$ —the treatment effect is the same *proportional* amount for everyone. When the goal is a percentage interpretation, the multiplicative assumption is arguably more natural: a 20% earnings loss means higher-earning workers lose more in levels, which is typically what we observe empirically.
- *Same estimand as Approaches 1 and 2:* With few controls, $\exp(\hat{\beta}_t^{Poisson}) - 1 \approx \hat{\beta}_t^{OLS} / \bar{y}$, so all three approaches estimate essentially the same percentage effect. Differences emerge when effects are large or when the outcome distribution has many zeros.

Earnings growth (Approach 4). This measures the individual-level percentage change $(y_{it} - y_{i,-1})/y_{i,-1}$, which is directly interpretable at the individual level. However, it is *undefined* when baseline earnings are zero, so it drops the extensive margin entirely. It can also be extremely noisy or ill-behaved when the denominator is small. This is a serious limitation whenever non-employment spells are prevalent.

Summary. When zeros are rare (as for main worker earnings before displacement), all four approaches should yield similar results. Differences emerge when zeros are prevalent—precisely the case for spousal outcomes and the added worker effect.

Figure 3: Expressing Earnings Losses in Percent (incl. zero earnings)



4 Added Worker Effect: Spousal Outcomes

This section estimates the **added worker effect** (AWE) — the change in spousal labor market outcomes in response to the primary worker’s job displacement.

We estimate event study regressions using the same treatment x relative-time dummies as for the main worker outcomes, but with spouse earnings, employment, and various transformations as dependent variables.

Note on zero baseline earnings: About 40% of spouses have zero earnings at baseline. This makes expressing the AWE in percentage terms considerably more challenging than for the displaced worker’s own earnings.

We present the following approaches:

- **Levels and Employment** (panels a–d): Raw means and event study estimates for spouse earnings in levels and employment. Well-defined for all observations.
- **Scale by Control Mean** (panels e–f): Raw mean difference and regression coefficient divided by the control group mean at each event time.
- **Poisson QMLE** (panel g): $\exp(\hat{\beta}_t) - 1$ handles zeros naturally.
- **Earnings Growth as % of HH Income** (panel h): $(y_{it}^S - y_{i,-1}^S)/(y_{i,-1} + y_{i,-1}^S)$.
- **Earnings Growth as % of Partner Earnings** (panel i): $(y_{it}^S - y_{i,-1}^S)/y_{i,-1}$.
- **Log with additive constant** (panels j–l): $\log(y^S + c)$ for $c \in \{0.01, 1, 100\}$.

Discussion: Percentage Interpretation for the Added Worker Effect

The zero-baseline problem. Earnings growth $(y_{it}^S - y_{i,-1}^S)/y_{i,-1}^S$ is undefined when baseline spouse earnings are zero. Conditioning on positive baseline earnings creates a selected sample: the extensive margin response—spouses *entering* the labor force in response to displacement—is missed entirely, even though it may be an important component of the AWE (Lundberg, 1985, *JPE*; Stephens, 2002, *JLE*). **Dividing by spouse baseline income does not work** for the full sample since it drops the approximately 40% of spouses who are not employed at baseline.

Scaling by the control mean. Dividing the level estimate by the control group mean works and includes all observations (zeros contribute to both numerator and denominator). However, when the control mean is small—as it is for initially non-employed spouses—small level changes can produce large and potentially misleading percentages.

Poisson QMLE targets the same parameter as scaling by the control mean. Like Approaches 1 and 2, Poisson QMLE uses spouse earnings in levels y_{it}^S as the outcome. With few controls, $\exp(\hat{\beta}_t^{Poisson}) - 1 \approx \hat{\beta}_t^{OLS}/\bar{y}^S$, so all three approaches estimate essentially the same percentage effect. The advantage of Poisson is that it handles zeros without any sample restriction or ad-hoc rescaling, gives a coherent percentage interpretation via $\exp(\hat{\beta}_t) - 1$, and captures both extensive and intensive margin responses in a single estimate. The multiplicative structure is especially appropriate here: when baseline spouse earnings are zero, the model implies zero expected contribution from the intensive margin, which is correct.

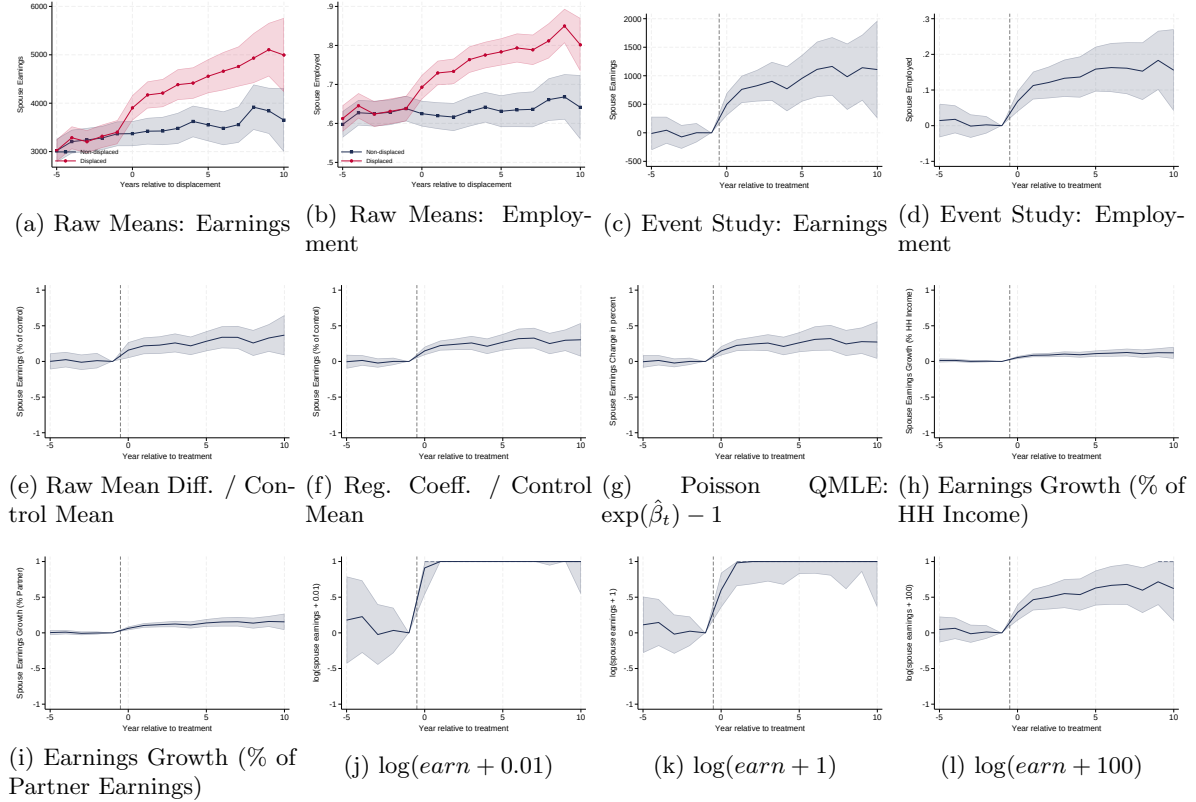
Earnings growth relative to household income. Normalizing by total household baseline earnings $(y_{i,-1} + y_{i,-1}^S)$ instead of spouse baseline earnings alone is always defined (since the primary worker has positive earnings by construction) and has a natural interpretation: it measures the spousal earnings response as a share of total household resources at risk.

Earnings growth relative to partner (displaced worker) income. Normalizing by the displaced worker’s baseline earnings $y_{i,-1}$ uses the same denominator as the displaced worker’s own earnings growth measure $(y_{it} - y_{i,-1})/y_{i,-1}$. This makes the two directly additive: the displaced worker’s earnings growth plus the spouse’s earnings growth (both normalized by the displaced worker’s pre-displacement earnings) gives the total household income change as a fraction of the

displaced worker's baseline earnings. This is always well-defined since displaced workers have positive baseline earnings by construction.

Log transformation with additive constant: $\log(y^S + c)$. A common approach to handle zero earnings is to estimate $\log(y + c)$ for some constant $c > 0$. However, Chen and Roth (*QJE*, 2024) show that the estimated treatment effect is highly sensitive to the choice of c and that there is no principled way to select it. We illustrate this with $c \in \{0.01, 1, 100\}$. The results vary dramatically across choices of c , confirming that this approach is unreliable for outcomes with many zeros.

Figure 4: Added Worker Effect: All Spouses

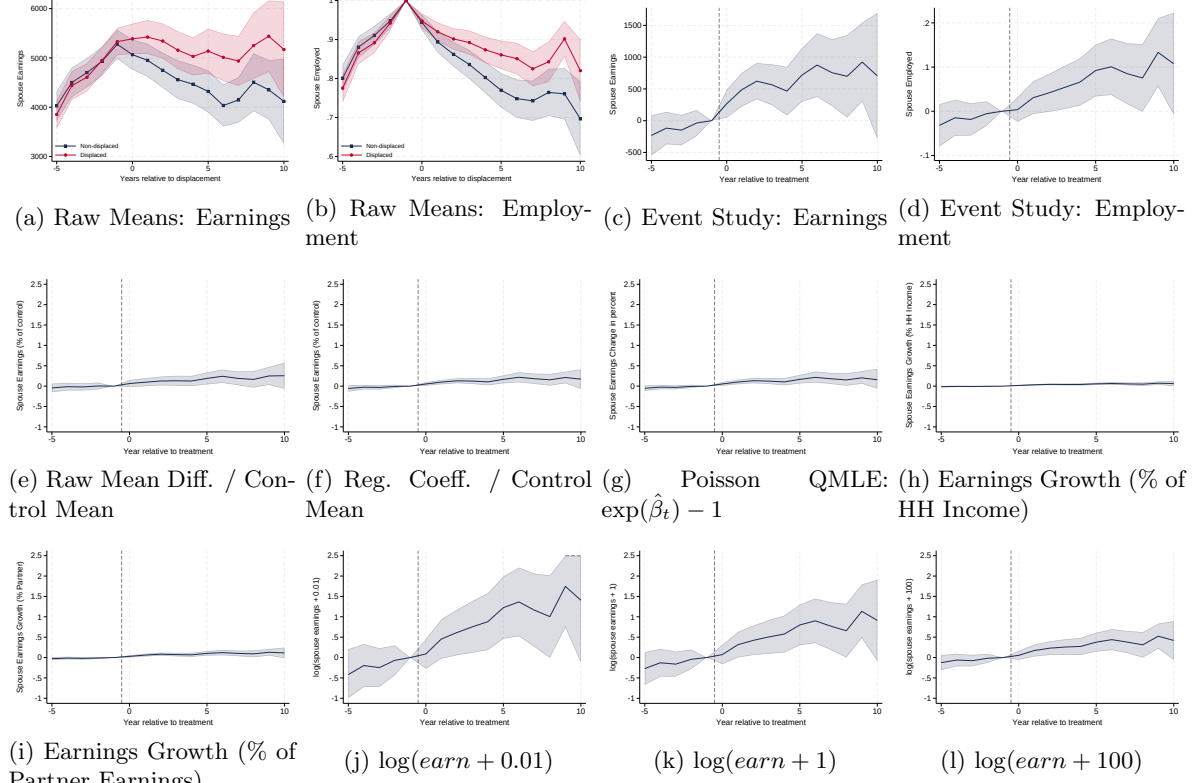


Notes: Panels (a)–(b) show raw means for displaced and non-displaced workers' spouses. Panels (c)–(d) show event study regression coefficients in levels. Panels (e)–(f) scale by the control group mean at each event time to express effects in percent. Panel (g) uses Poisson QMLE, which handles zeros naturally and gives a direct percentage interpretation. Panel (h) normalizes spouse earnings changes by total household baseline income. Panel (i) normalizes by the displaced worker's baseline earnings—same denominator as the displaced worker's own earnings growth, so the two are directly additive. Panels (j)–(l) illustrate that $\log(y + c)$ is highly sensitive to the choice of c (Chen and Roth, *QJE*, 2024).

5 AWE for Spouses Employed at Baseline

This section shows the added worker effect for spouses who were employed at baseline. Since these spouses have positive baseline earnings, all measures including earnings growth are well defined.

Figure 5: AWE for Baseline Employed Spouses

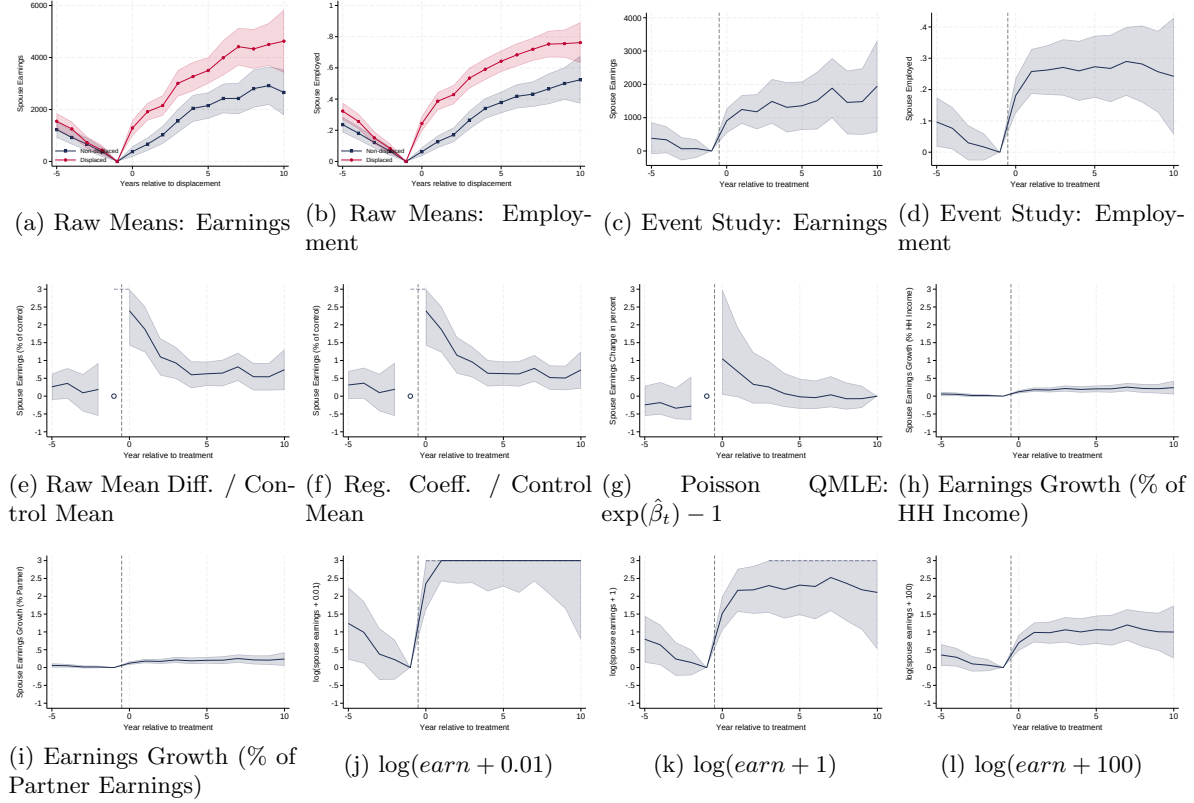


Notes: Conditional on spouse being employed at baseline. All percentage measures are well-defined since baseline spouse earnings are positive. Panels (j)–(l) show that $\log(y + c)$ estimates vary with c even in this subsample (Chen and Roth, *QJE*, 2024).

6 AWE for Spouses Not Employed at Baseline

This section shows the added worker effect for spouses who were not employed at baseline. Since baseline spouse earnings are zero, earnings growth $(y_{it}^S - y_{i,-1}^S)/y_{i,-1}^S$ is undefined. Dividing by spouse baseline income does not work here since it requires dividing by zero. However, normalizing by household income or by the displaced worker's baseline earnings remains well-defined, as does Poisson QMLE.

Figure 6: AWE for Baseline Non-Employed Spouses



Notes: Conditional on spouse being *not* employed at baseline. Panels (a)–(d) are in levels and well-defined. Panels (e)–(g) use hollow circles at the omitted category. Scaled outcomes are undefined (division by zero) and Poisson coefficients are not identified for groups where outcomes are precise zeros. Panels (h)–(i) normalize by the displaced worker's (positive) baseline earnings or total household income and are well-defined even for baseline non-employed spouses. Panels (j)–(l) illustrate the sensitivity of $\log(y + c)$ to the choice of c (Chen and Roth, *QJE*, 2024).

References

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